where in the third line we use the independence assumption and the fact that X' and Y' have zero mean.

Observe that $\kappa_\psi(\rho) = (1-r^2)\kappa(\rho)$, or $\kappa'_\psi(\rho) = (1-r^2)\kappa'(\rho) + r^2$. Now, let us consider the conditions for the four cases of Theorem 1. The decision boundaries are if $\kappa(0)$ is positive or not, if $\kappa'(1)$ is above, equal to, or below 1. Now, note that for $r \in (0,1)$, strict positivity of $\kappa_\psi(0) = (1-r^2)\kappa(0)$ remains the same as $\kappa(0)$. Furthermore, $\kappa'_\psi(1)$ is a weighted average of $\kappa'(1)$ and 1. Thus, for all values $r \in (0,1)$, it holds $\kappa'_\psi(1)$ is above, equal to, or below 1, if and only if $\kappa'(1)$ has the corresponding property. Now, we can turn our focus on the convergence rates. For this, we can focus on α in each one of the four cases.

- If $\kappa(0) = 0$ then $\kappa_\psi(0) = r^2\kappa(0)$ and we have $\alpha = 1/(2-\kappa'_\psi(0))$. Now, note that $\kappa'_\psi(0) = (1-r^2)\kappa'(0)+r^2$ is a weighted average between $\kappa'(0) < 1$ and 1, and thus, the larger the value of r, the larger α would be, and the slower the convergence.
- If $\kappa(0) > 0$ and $\kappa'(1) < 1$ then we have $\kappa_\psi(0) = 0$ and $\kappa'_\psi(1) < 1$, and $\alpha = \kappa'_\psi(1) = (1-r^2)\kappa'(1) + r^2$. Thus, the larger the residual, the closer α becomes to 1, and the slower the convergence will be.
- If $\kappa(0) > 0$ and $\kappa'(1) = 1$, then we have the same for κ_ψ, and

$$\begin{aligned}\alpha &= 1 - \kappa_\psi(0) - \kappa'_\psi(0)\\ &= 1 - (1-r^2)\kappa(0) - (1-r^2)\kappa'(0) - r^2\\ &= (1-r^2)(1-\kappa(0)-\kappa'(0))\end{aligned}$$

  Now, recall that we have $\kappa(0) = c_0^2$ and $\kappa'(0) = c_1^2$, and thus we have $1-\kappa(0)-\kappa'(0) = \sum_{k=2}^\infty$, which is necessarily positive for a nonlinear activation:

$$\alpha = (1-r^2)\sum_{k=2}^\infty c_k^2.$$

  We can now see that for larger r, α will be smaller, which in this case implies a slower convergence.
- If $\kappa(0) > 0$ and $\kappa'(1) > 1$ then same holds for κ_ψ, and the convergence rate α .

□

Proof of Proposition 3. Recall the assumption that ϕ obeys $\mathbb{E}\,\phi(X)^2 = 1$, $X \sim N(0,1)$. Let us assume that $Z \sim N(0, I_d)$ represents the Gaussian pre-activations from the previous layer We can consider the joint normalization and activation layer ψ in the mean-field regime:

- $\psi = \phi \circ RN$ and $\psi = \phi \circ LN$: Because each element of Z has zero mean and unit variance, in the mean field regime, the sample mean and variances will be equal to the their population counterparts, implying that $LN(Z) = Z$ and $RN(Z) = Z$. In other words, they act as identity. Thus, in both cases, in the mean-field regime we have $\psi = \phi$. Thus, the kernel map also remains the same.
- $\psi = RN \circ \phi$: In this case, because of the assumption on activation for Gaussian preactivations we have $\mathbb{E}\,\phi(Z_i)^2 = 1$, for all $i = 1, \ldots, d$. Because elements of Z are i.i.d., by law of large numbers $\frac{1}{d}\|\phi(Z)\|^2$ will converge to its expected value 1. Thus, again, the normalization step in RN becomes ineffective, implying that in the mean-field we have $\psi = \phi$. Thus, the kernel map also remains the same.
- $\psi = LN \circ \phi$: By definition of kernel map of activation, we have

$$\mathbb{E}\,\phi(Z_i) = \sqrt{\kappa_\phi(0)}, \qquad \mathrm{Var}(Z_i) = \mathbb{E}\,\phi(Z_i)^2 - (\mathbb{E}\,\phi(Z_i))^2 = 1 - \kappa_\phi(0)$$

  Thus, again by law of large numbers we will have

$$\psi = (\phi - \sqrt{\kappa_\phi(0)})/\sqrt{1-\kappa_\phi(0)}.$$

Now, if we look at the kernel map of this affine transformation, we have

$$\begin{aligned}\kappa_\psi(\rho) &= \mathbb{E}\,\psi(X)\psi(Y)\\ &= \mathbb{E}\,\frac{(\phi(X)-\sqrt{\kappa_\phi(0)})}{\sqrt{\kappa(1)-\kappa(0)}}\frac{(\phi(Y)-\sqrt{\kappa_\phi(0)})}{\sqrt{1-\kappa_\phi(0)}}\\ &= \frac{\mathbb{E}\,\phi(X)\phi(Y)-\kappa_\phi(0)}{1-\kappa_\phi(0)}\\ &= \frac{\kappa_\phi(\rho)-\kappa_\phi(0)}{1-\kappa_\phi(0)}\end{aligned}$$

which concludes the proof.

□

Remark S.1. *Another interesting implication is that if we take a look at discrete and continuous kernel dynamics for a very strong residual (when $r \to 0$), the kernel ODE becomes exact. Let us denote the Hermite coefficients of the residual kernel by $\tilde{c}_k$. For the first term, we have $\tilde{c}_1^2 = r^2c_1^2 + (1-r^2)1$ and for all other terms $\tilde{c}_k^2 = r^2c_k$. Note that as $r \to 0$, we have $\tilde{c}_1^2 \to 1$, and for all other terms $\tilde{c}_k^2 \to 0$. Thus, in the discrete kernel dynamics, the right-hand side will converge to zero. This implies that in the limit of very strong residuals, the kernel ODE gives the exact solution to the kernel dynamics.*

Remark S.2. *Suppose there is an activation ϕ with a negative fixed point $\rho^* < 0$. Let n be an integer such that $\rho^* < -1/(n-1)$. Let $x_1, \ldots, x_n \in \mathbb{R}^d$ be vectors that have non-zero inner products. Construct an MLP with activation ϕ and let $y_1, \ldots, y_n$ be the output of this MLP. Thus, if the depth is sufficiently large, the pairwise similarity between each pair will converge to ρ^*. Thus, their output Gram matrix $G = [\langle y_i, y_j\rangle]_{i,j\le n}$ has unit diagonals and off diagonals equal to ρ^*. By analyzing the eigenvalues of G, we find: The top eigenvalue corresponding to the all-ones eigenvector is $\lambda_1 = 1 + (n-1)\rho^*$. Because we assumed $\rho^* < -1/(n-1)$,, we have $1 + (n-1)\rho^* < 0$, which is a contradiction. Because G by construction must be positive and semi-definite.*

Corollary S.1. *If Hermite coefficients of ϕ has Hermite expansion $\phi = \sum_{k=m}^{\infty} c_k \mathrm{he}_k$, then kernel sequence of MLP with activation ϕ converges to zero with double exponential rate*

$$|\rho_\ell| \le |\rho_0|^{m^\ell}.$$

The proof of corollary is by simply observing that, the slowest convergence happens when the weight of coefficients is concentrated at $k = m$ term, and for that case, we can observe that $\|\kappa(\rho)\| \le \|\rho\|^m$, and by induction over ℓ we can prove the claim.

B Validation of the global convergence theorem

Here we will provide some numerical validation of the global convergence theorem.

Remark S.3. *First off, note that Theorem 1 requires that the activation functions preserve (do not increase or decrease) the energy of the pre-activations, $\mathbb{E}\,[\phi(X)^2] = 1$, for $X \sim N(0,1)$. While some activation functions, namely* `SeLU` *Klambauer et al. [2017], have this property, we can achieve it for all activation functions by inversely scaling them by a constant factor, equal to $C = \sqrt{\mathbb{E}\,[\phi(X)^2]}$. After this step, we can quantify various values relevant to Theorem 1. This step is applied for both the results in the table, as well as Figures S.1 and S.2. The scaling constant C for each activation function is shown in Table S.1.*

Recall the conditions for each convergence from Theorem 1:

- $\kappa(0) = 0$: Case 1, exponential convergence towards $\rho^* = 0$.
- $\kappa(0) > 0$:
  - $\kappa'(1) < 1$: Case 2, exponential convergence towards $\rho^* = 1$.
  - $\kappa'(1) = 1$: Case 3, polynomial convergence towards $\rho^* = 1$.
  - $\kappa'(1) > 1$: Case 4: exponential convergence towards some $\rho^* \in (0,1)$.

In Table S.1, we have quantified the fixed points, relevant quantities, and convergence rates for each activation function. This table shows that for the most popular activation functions that are used in practice, we can quantify a fixed point and explicit rate that their kernel sequence will converge to.

It is worth noting that because Theorem 1 provides an upper bound, we cannot directly compare the quantified value of α for different cases, particularly when they correspond to different cases. In other words, because this is a worst-case bound, it is possible that an activation function manifests a much higher convergence than predicted. We will further explore the gaps between the empirical values and the upper bound in the subsequent figures.

ϕ	C	α	$\rho_\star$	$\kappa(\rho_\star)$	$\kappa(0)$	$\kappa'(0)$	$\kappa'(1)$	$\kappa'(\rho^*)$	Convergence
`tanh`	0.63	0.93	0.00	0.00	0.00	0.93	1.18	0.93	case 1 /Exp.
`SeLU`	1.00	0.97	0.00	0.00	0.00	0.97	1.06	0.97	case 1 /Exp.
`ReLU`	0.71	0.95	1.00	1.00	0.32	0.50	0.95	0.95	case 2/Exp.
`sigmoid`	0.54	0.15	1.00	1.00	0.85	0.15	0.15	0.15	case 2 /Exp.
`exp`	2.72	0.74	1.00	1.00	0.37	0.37	1.00	1.00	case 3 /Poly.
`GELU`	0.65	0.93	0.76	0.76	0.19	0.59	1.07	0.93	case 4 /Exp.
`CELU`	0.80	0.97	0.60	0.60	0.04	0.90	1.04	0.97	case 4 /Exp.
`ELU`	0.80	0.97	0.60	0.60	0.04	0.90	1.04	0.97	case 4 /Exp.

Table S.1: Activation functions: a review of most commonly used activation functions (normalized, see Remark S.3), according to Theorem 1. In the last column, we have noted the case of convergence according to Theorem 1, as well as the speed of convergence (Poly: polynomial, Exp: exponential, not to be confused with activation $\phi = \texttt{exp}$) in depth. Note that only the `exp` activation function has polynomial convergence.

B.1 Figures S.1 and S.2

Each row of the figures shows one activation function and each column is dedicated to the following (from left to right). The First column shows the activation function itself, up to some scaling (see Remark S.3). In the following, we will describe the last three columns.

Kernel map and fixed point iterations The second column shows the kernel map κ of each activation function (blue), as defined in Definition 1, and the fixed iterations over this kernel (red), as defined by pairs of points

$$(\rho_{\ell+1}, \rho_\ell), \qquad \rho_{\ell+1} := \kappa(\rho_\ell)$$

where ρ_0 indicates the initial value that is arbitrarily chosen (shown as a red dot marker). As the iterations progress, they converge to the fixed point (ρ^*, red star marker). The identity map is shown to demonstrate visually how the iterations lead to the fixed point.

It is worth noting that kernel maps of all activation functions are analytic, i.e., smooth up to an arbitrary degree, even when the activation function itself is non-smooth, which is the case for `ReLU` and `SeLU` at $x = 0$. We can also see that despite the non-monotonicity of some activation functions, such as `GELU`, the kernel map maintains its unique properties, such as having a unique globally attracting fixed point.

These kernel map plots give important insight into the global convergence of the kernel and the predictions of Theorem 1. Intuitively, the fixed point is where the kernel map intersects with the identity line, and the speed at which it converges to the fixed point is inversely related to the slope at ρ^*. For example, `tanh` and `SeLU` are close to the identity line, and they exhibit very slow convergence, while `sigmoid` deviates the most from the identity and shows the fastest convergence rate. Overall, we can see here why deviation from the identity, as captured by $\kappa'(\rho^*)$, and other terms, play an important role in the convergence of the kernel sequence.

It is worth noting that while `sigmoid` and `tanh` are tightly related by the formula $\texttt{tanh}(x) = 2 \cdot \texttt{sigmoid}(2x) - 1$, their convergence properties are drastically different, with `tanh` converging exponentially towards zero (orthogonality or independence bias), while `sigmoid` converges exponentially towards 1 (strong similarity bias). We can explain this by observing that the shifting ensures that the activation function has zero-mean postactivations, making ρ^* a fixed point.

Kernel convergence theory vs empirical results. The third and fourth columns show the kernel sequence (empirical: blue, theory: red) as a function of depth ℓ. The theoretical bound corresponds to Theorem 1. In the fourth column, we